

IJMI Machine Learning Reporting Checklist

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This checklist maps the reported study to key reporting domains in TRIPOD+AI and PROBAST-AI. It is intended as a submission supplement; item wording is summarized rather than reproduced.

Domain	TRIPOD+AI PROBAST-AI topic	/ Study location and status
Study design	Study objective and design	Introduction; retrospective next-visit, four-class prediction task.
Data source	Data provenance and setting	Materials and Methods, Dataset; NACC Uniform Data Set and participating centers.
Participants	Eligibility and cohort construction	Dataset; consecutive visits and exclusion of missing targets are described.
Outcome	Outcome definition and prediction horizon	Target Variable Definition; CDRGLOB at visit $t + 1$, mapped to four classes.
Predictors	Predictor availability and definition	Feature Engineering; 101 predictors after excluding identifiers and future target.
Preprocessing	Missing data and feature processing	Data Preprocessing; median imputation, encoding, and finite-value handling.

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Domain		TRIPOD+AI PROBAST-AI topic	/ Study location and status
Model development		Algorithms and hyperparameters	Gradient Boosting Framework; base XGBoost and ablation configurations are specified.
Data splitting		Leakage prevention and resampling	GroupShuffleSplit at patient level; SMOTE restricted to ablation training data.
Validation		Internal validation strategy	Single patient-level 80/20 train-test split with a 20% internal validation holdout for early stopping.
Performance		Discrimination and classification metrics	Results; accuracy, macro/weighted F1, class metrics, ROC curves, and error analysis.
Calibration		Calibration reporting	Model Calibration and Confidence Analysis; class-specific ECE and reliability curves are reported.
Subgroups		Fairness and bias assessment	Demographic Fairness Analysis; sex, age, race, and explicit Unknown subgroup results.
Interpretability		Explainability and feature attribution	SHAP TreeExplainer analysis and class-specific feature rankings.
Reproducibility		Code, data access, and model specification	Data Availability Statement, repository, notebook, requirements, and supplementary outputs.

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Intended use	Clinical context and limitations		Clinical Roadmap; primary-care/community screening pathway and prospective validation plan.
Risk of bias	Applicability and limitations		Limitations; cohort representativeness, subgroup sizes, time horizon, feature availability, and oversampling limitations.

Checklist notes

The study is a retrospective single-split evaluation and is not a prospective impact or randomized implementation study. Calibration, subgroup performance, and external transportability require prospective and multi-site confirmation before clinical deployment. The intended use is decision support and risk stratification, not autonomous diagnosis or treatment selection.